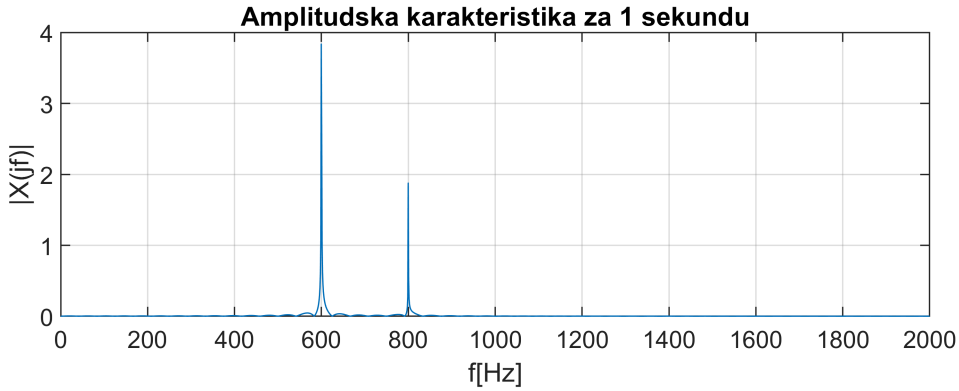
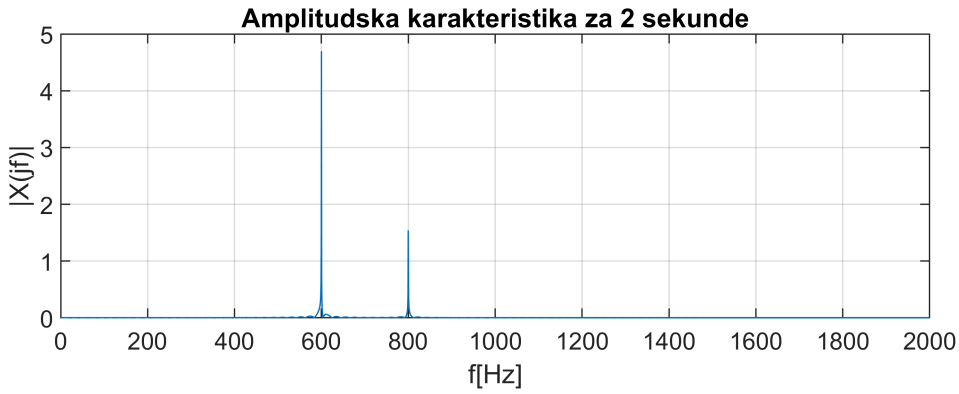
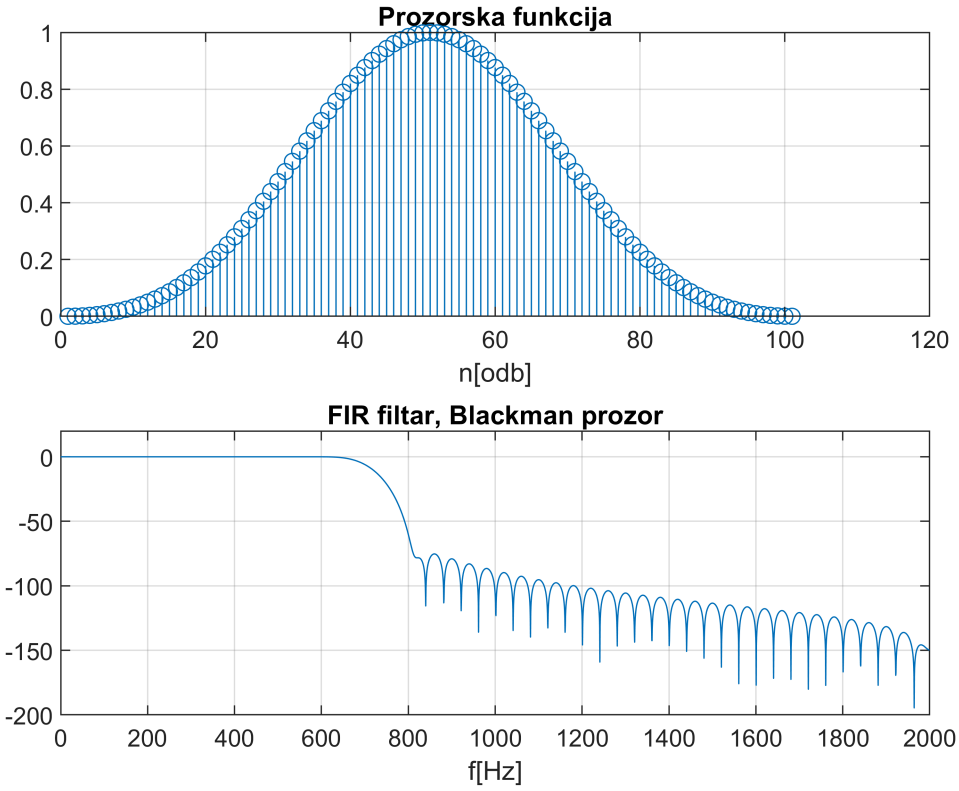
```
clear all; close all; clc;

Fs = 4000;

t = 0 : 1/Fs : 2;
x = 5*sin(2*pi*600*t) + 2*cos(2*pi*800*t);
N = 2^nextpow2(length(x));
f1 = 0 : Fs/N : Fs/2;
X = fft(x,N)/length(x);
X1 = abs(X(1 : N/2+1));
X1(2 : N/2+1) = 2*X1(2 : N/2+1);

t2 = 0 : 1/Fs : 1;
x2 = 5*sin(2*pi*600*t2) + 2*cos(2*pi*800*t2);
N2 = 2^nextpow2(length(x2));
f2 = 0 : Fs/N2 : Fs/2;
X2 = fft(x2,N2)/length(x2);
X3 = abs(X2(1 : N2/2+1));
X3(2 : N2/2+1) = 2*X3(2 : N2/2+1);

figure(1);
subplot(2, 1, 1);
plot(f2, X3);
xlabel('f[Hz]'); ylabel('|X(jf)|'); grid on;
title('Amplitudska karakteristika za 1 sekundu');
subplot(2, 1, 2);
plot(f1, X1);
xlabel('f[Hz]'); ylabel('|X(jf)|'); grid on;
title('Amplitudska karakteristika za 2 sekunde');
```

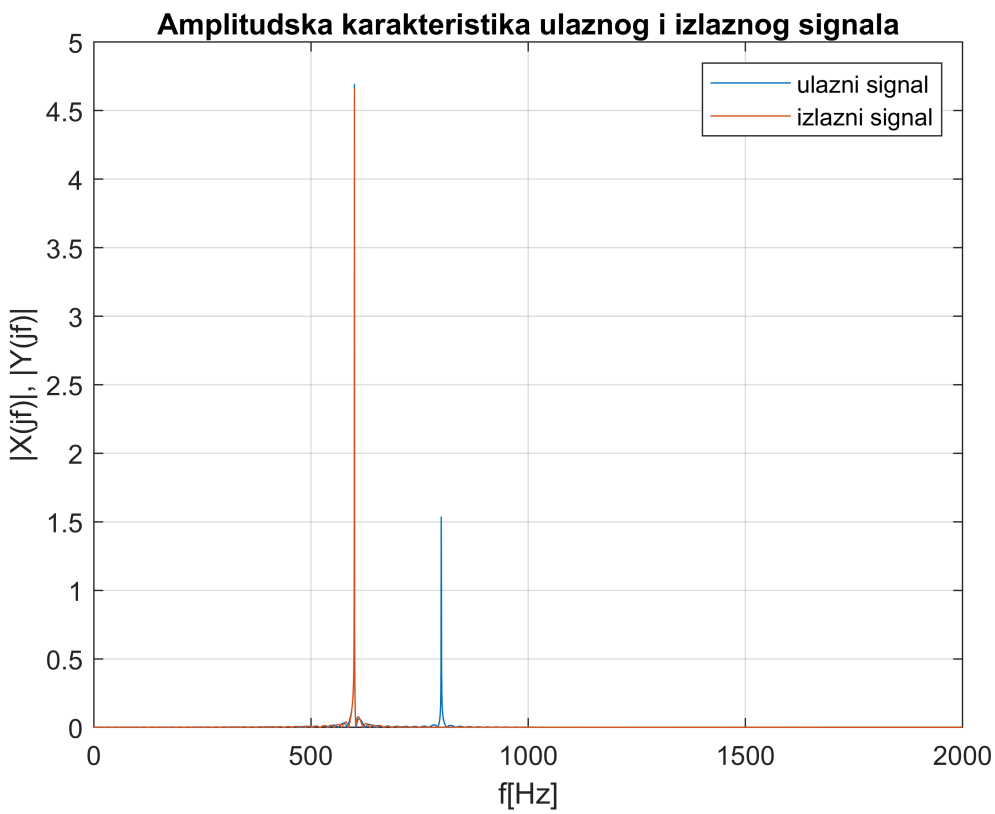




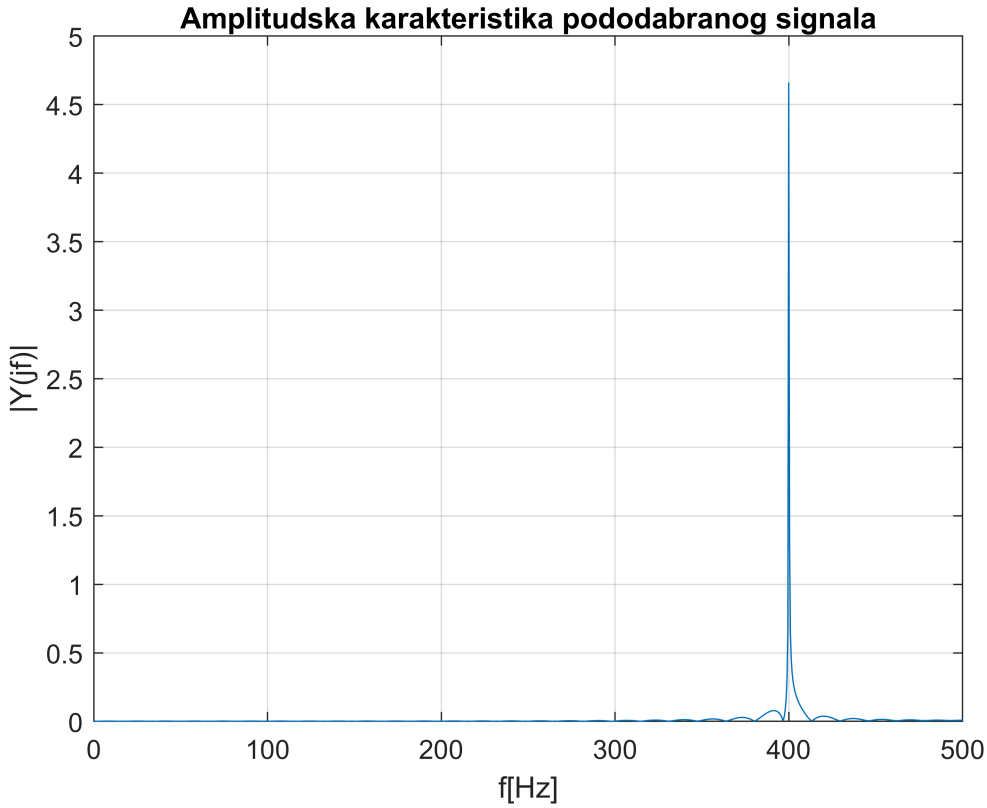

```
%Sto je veca duzina signala, spektar je sve tacniji i uzi oko
%pravih ucestanosti, suprotno za manju duzinu je manje informativan
%--------------------------------------------------
n = 100;
window = blackman(n+1);
Wn = 700/(Fs/2);
b = fir1(n, Wn, window);
a = 1;
[hz, fz] = freqz(b, a, N/2+1, Fs);

figure(2);
subplot(2, 1, 1);
stem(window);
xlabel('n[odb]'); grid on;
title('Prozorska funkcija');
subplot(2, 1, 2);
plot(fz, 20*log10(abs(hz)));
xlabel('f[Hz]'); ylim([-200 20]); grid on;
title('FIR filtar, Blackman prozor');
```




```
y = filter(b, a, x); %filtriranje
Y = fft(y, N)/length(y);
Y1 = abs(Y(1 : N/2+1));
Y1(2 : N/2+1) = 2*Y1(2 : N/2+1);

figure(3);
plot(f1, X1); hold on;
plot(f1, Y1);
xlabel('f[Hz]'); ylabel('|X(jf)|, |Y(jf)|'); grid on;
legend('ulazni signal', 'izlazni signal');
title('Amplitudska karakteristika ulaznog i izlaznog signala');
```




```
%sound(y, Fs);
%-------------------------------------------------------------
% Kosi delovi na jelkici:
% [2000 4000] za nn = 1, 600 ne upada, nema zamene ucestanosti
% [1000 2000] za nn = 2, 600 ne upada, nema zamene ucestanosti
% [500 1000] za nn = 4, 600 upada, ima zamene ucestanosti, 600 -> 400

nn = 4;
Fsn = Fs/nn;
z = y(1 : nn : length(y));

Nn = 2^nextpow2(length(z));
f1n = 0 : Fsn/Nn : Fsn/2;
Yn = fft(z,Nn)/length(z);
Y1n = abs(Yn(1 : Nn/2+1));
Y1n(2 : Nn/2+1) = 2*Y1n(2 : Nn/2+1);

figure()
plot(f1n, Y1n);
xlabel('f[Hz]'); ylabel('|Y(jf)|'); grid on;
title('Amplitudska karakteristika pododabranog signala');
```


Amplitudska karakteristika pododabranog signala
|Y(jf)|
f[Hz]
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
0
100
200
300
400
500